

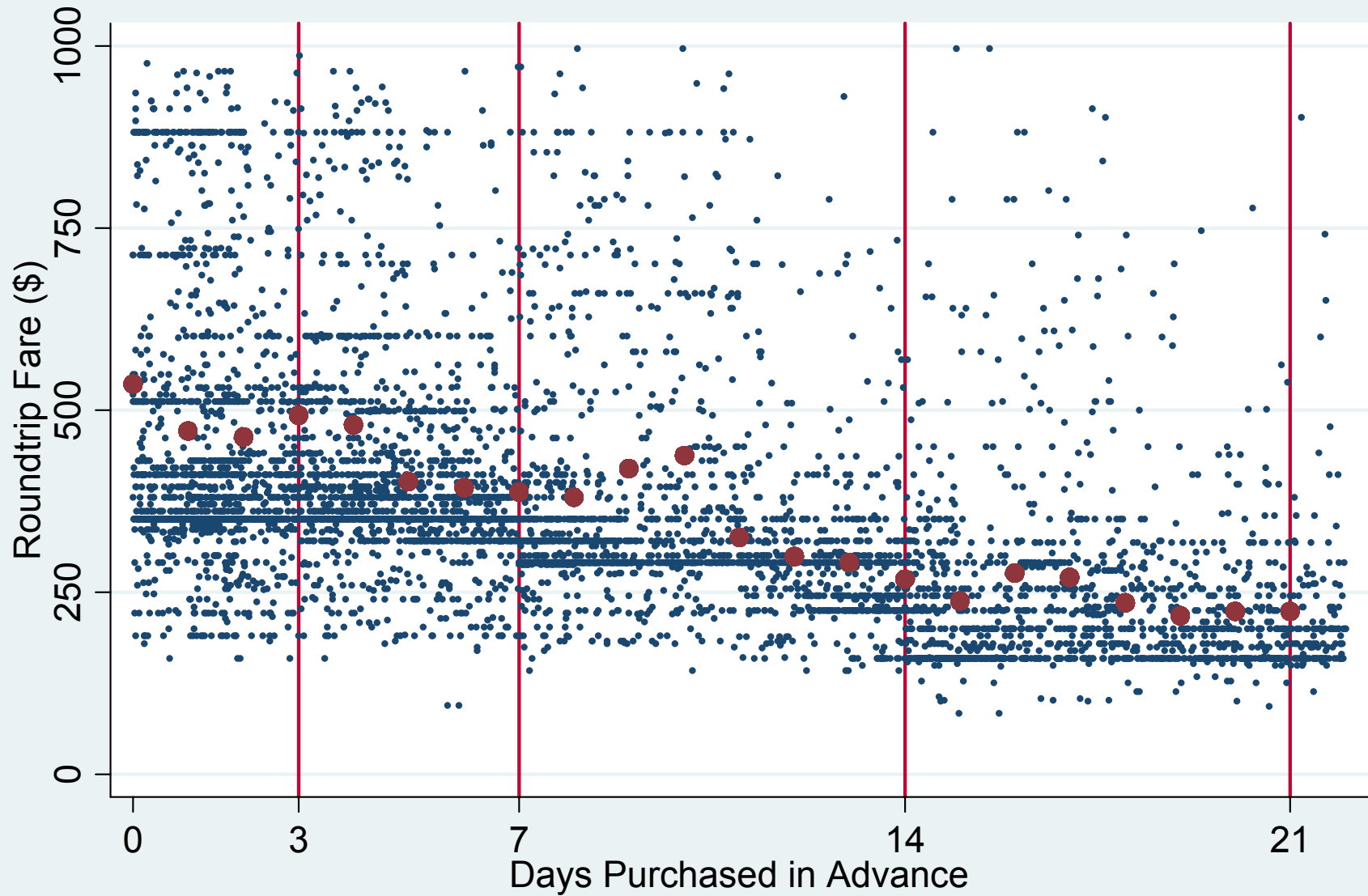
The Effect of Competition on Price Dispersion in Airlines

Papers:

Borenstein, Severin and Nancy Rose.
“Competition and Price Dispersion in the U.S.
Airline Industry,” *Journal of Political Economy*,
102(4):653-683, August 1994.

Gerardi, Kristopher S. and Adam Hale Shapiro.
2009. “Does Competition Reduce Price
Dispersion? New Evidence from the Airline
Industry”, *Journal of Political Economy*, 117(1): 1-
37, February 2009.

American: DFW-LAX Tickets Sold for Travel in 2004Q4

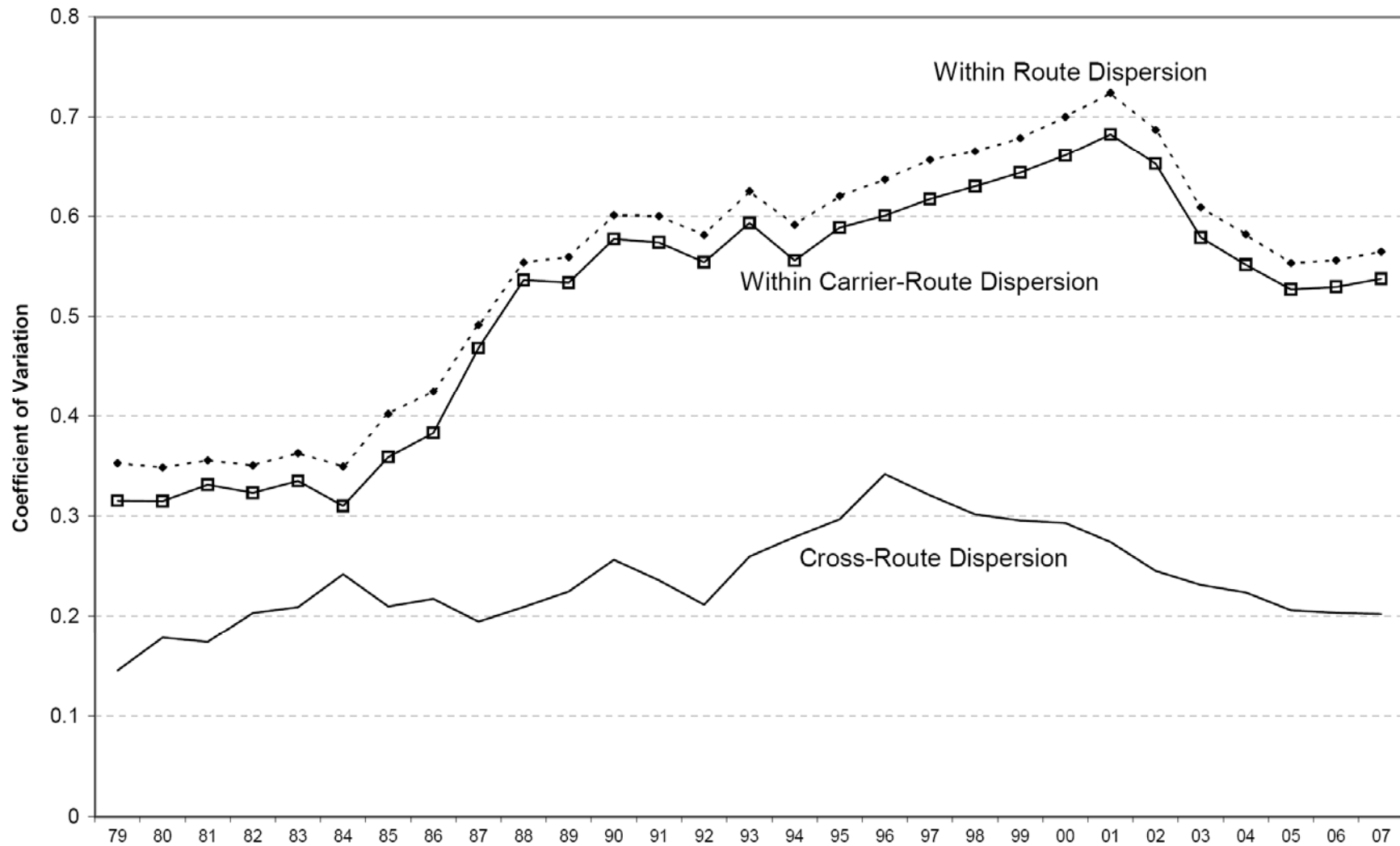


• Roundtrip Fare • Average Fare

Fare Dispersion on a Route

- 1986: Gini coefficient of 0.18
 - → two randomly selected tickets have an expected price difference = 36% of the mean
- 2004: (Separate dataset – Puller, Sengupta, Wiggins)
 - Gini for major carriers about 0.28 on a carrier-route (on a set of routes that are larger than the set in BR)

Figure 5: Within-Route and Cross-Route Price Dispersion, 1979-2007



Source: Borenstein & Rose (2008)

Gini Variation Across Routes

- Contribution of bottom half of passengers
 - 10th pctl route: 37%
 - 90th pctl route: 27%
 - 100th pctl route: 20%
- $\text{Corr}(\text{HHI}, \text{Gini}) = -0.153$

Research Questions in these Papers

- Does price dispersion vary with competition of the route?
 - Overall goal to address: Is price dispersion due to price discrimination OR differences in cost?
- Gerardi & Shapiro: does competition reduce price discrimination?
- Neither really address welfare consequences
 - In fact, price discrimination can $\uparrow$ or $\downarrow$ welfare
 - So normative implications are not clear

Institutionally, How Are Fares Chosen?

- “Yield management” (operations research)
 - *Planning* department chooses flight schedule & fare structure.
 - *Yield management* dept chooses seat allocated to each fare
- ▶ The “Capacity allocation” problem
 - Seat allocation to different fare classes (or sets of restrictions)
 - Given pre-determined fares, how many seats to allocated to each fare bucket given estimates of (possibly dependent) demands by each type
 - balance the tradeoffs when increasing low fare seats by one unit: spoilage (a seat won’t be filled) vs. dilution (will offset a full fare purchase)

Interpretation

- Why not just “ask the airlines” how they price?
- Actual airline decision is a complex OR problem
 - “To solve the system-wide yield-management problem would require approximately 250 million decision variables. Because this mathematical programming formulation is intractable, American Airlines Decision Technologies has developed a series of operations research models. These models effectively reduce the large problem to three much smaller and far more manageable subproblems while still realistically modeling the real-world situation.”
 - Barry Smith, *Interfaces*, 1992
- Economic models as reduced-form interpretation

2 Models Yielding Price Dispersion

1. “Standard textbook model”

- Textbook 3rd degree price discrimination requires market power
- $P > MC$ cannot be sustained absent mkt pwr
- More competitive → less price discrimination → less price dispersion

2. Price discrimination can increase moving from monopoly to imperfect competition

- Intuition: Suppose monop. sets 2 fares: “High” to less inelastic business travelers, “low” to elastic tourists
- Entry induces both “high” and “low” to fall, but “low” falls by more

Data Available Drives the Empirical Strategy

- (Close to) ideal data:
 - ticket-level data on ticketing restrictions, frequent flyer membership, (shadow) cost of a seat (see basis codes)
- Standard dataset for airlines
 - DOT's Origin & Destination Survey
 - (see http://www.bts.gov/programs/airline_information/)
 - 10% sample of tickets, but strip of information on ticket characteristics (refundability, travel restrictions, advance purchase restriction, date of purchase, how full the itinerary's flights are...)
 - So basically know that somebody flew for airport A to airport B on carrier X in the 2nd quarter of 1986 and paid \$Y
 - Therefore, must exploit cross-market variation, and find predictions about what types of markets yield certain types of price discrimination
- See Puller, Sengupta, Wiggins (2012) for a much cooler dataset 😊

Basis Codes

FARES LAST UPDATED 21JUL 2:37 PM
 >\$DSMFORD13OCT-NLX+UA
 UA DEPART 13OCT AIRPORT FARE

PUBLIC FARES
 U.S. PASSENGER FACILITY CHARGES / SURCHARGES MAY APPLY
 TAXES AND FEES MAY VARY DEPENDING ON THE BOOKED ITINERARY

	CX	USD FARE	FARE BASIS	AP	MIN/ MAX	XL FE	TVL DATES FIRST/LAST	TKT DATES FIRST/LAST
SMFCHI								
1	UA	99.00	LA14ON8	14+	--/--	++	18AUG/18NOVC	-/-
2	UA	139.00	TA14ON9	14+	--/--	++	18AUG/18NOVC	-/-
3	UA	190.00	SA14NN	14+	--/--	++	-/08JANC	-/-
4	UA	228.00	WA14KN	14+	--/--	++	-/-	-/-
5	UA	246.00	WA10KN	10+	--/--	++	-/-	-/-
6	UA	275.00	VA7KN	07+	--/--	++	-/-	-/-
7	UA	310.00	QA7KN	07+	--/--	++	-/-	-/-
8	UA	335.00	HA7KN	07+	--/--	++	-/-	-/-
9	UA	365.00	UA7KN	07+	--/--	++	-/-	-/-
10	UA	409.00	UA0KY	+	--/--	--	-/-	-/-
11	UA	509.00	EA0KY	+	--/--	--	-/-	-/-
12	UA	709.00	MA0KY	+	--/--	--	-/-	-/-
13	UA	1062.00	BUA	--	--/--	--	-/-	-/-
14	UA	1081.00	HUAUP	--	--/--	--	-/-	-/-
15	UA	1269.00	YUAS	--	--/--	--	-/-	-/-
16	UA	1284.00	BUAUP	--	--/--	--	-/-	-/-
17	UA	1494.00	FUAS	--	--/--	--	-/-	-/-
18	UA	1727.00	Y	--	--/--	--	-/-	-/-
19	UA	2674.00	F	--	--/--	--	-/-	-/-

- **The first letter:**
 - First Class Examples: A, F, P
 - Business Class Examples: C, D, J
 - Economy Class Examples: Y, B, H, K, L, M, O, N, S, V, Q & others.
- **The second letter:**
 - L: low season
 - H: high season
 - J: mid season
- **The 3rd and 4th characters:**
 - 7 / 14/ 21: 7-day / 14-day / 21 day advance;
 - Example: ME721NR (class 'M', 21 days in advance, non-refundable.)
 - X: midweek;
 - W: weekend;
- **Other restrictions:**
 - N in the second letter sometimes denotes 'no fuel surcharges';
 - L in the second letter sometimes refers to 'low season';
 - S at the end sometimes denotes 'Sunday night';
 - NR at the end denotes 'non-refundable';

More on model (2)

- 2 relevant elasticities for each customer type
 1. “industry” elasticity – demand for product of air travel
 - This “monopoly-type” p.d. implies more competitive → less dispersion
 2. “cross-elasticity” – demand across brands/airlines
 - This “competitive-type” p.d. implies more competitive → more dispersion
- See Stole (HIO, Vol 4), p. 2235 for details.
- See Borenstein (1985) and Holmes (1989)
- Simulated via a “clock” model...
- Empirical Question: does “monopoly-type” or “competitive-type” p.d. dominate?

Classic Reduced-form Approach

TABLE 1

PREDICTED COEFFICIENT SIGNS UNDER ALTERNATIVE MODELS OF PRICE DISPERSION

VARIABLE	PRICE DISCRIMINATION			PEAK-LOAD PRICING: Systematic and Stochastic (4)
	Monopoly Markets: Monopoly- Type (1)	Nonmonopoly Markets		
		Monopoly- Type (2)	Competitive- Type (3)	
Concentration (FLTHERF)	+	+	—	?
Market density (FLTTOT)	+	—	—	?
Market share (FLTSHARE)	+	+	—	?
Endpoint dominance (ENDDOMO)	+	+	+	0
Population homogeneity (TOURIST)	—	—	—	?
Variation in shadow flight ca- pacity cost (SDCAPFLT)	0	0	0	+
Variation in shadow airport capacity cost (SDCAPAPT)	0	0	0	+
Uncongested airport (DUMAPT)	0	0	0	—

Comparative Statics for P.D.

- Note “flavor”: tradeoff of taking institutional details seriously vs. making theory simple/elegant
- More consumer heterogeneity $\uparrow$ p.d. for both “c” and “m” type
 - Desired: tourist/business mix
 - Proxy: TOURIST. As rises, non-monot relationship. But B&R say $\uparrow$ TOURIST $\rightarrow$ $\downarrow$ dispersion
 - TOURIST is $f(\text{data on hotel income} / \text{total personal income})$
- More dense routes/frequent flights
 - “c” type: more cross-carrier substitution $\rightarrow$ more cross-elasticity $\rightarrow$ less dispersion
 - “m” type: more flights $\rightarrow$ increase “value” $\rightarrow$ decrease market elasticity $\rightarrow$ more dispersion
 - FLTOT = total # flights on a route

Comparative Statics (contd)

- Dominant firm on a route
 - “c” type: Dominant airline (e.g. 10 of 11 flights) less responsive to cross-elasticities → less dispersion
 - “m” type: simulations show share → more dispersion. ?intuition?
 - Continuous: FLTSHARE
 - Discrete: LARGE-DUOP (large in duop) and SMALL-DUOP (small in duop)
- Frequent Flyer Programs
 - Reduce both industry (increase reservation value) and cross-elasticity (brand loyalty)
 - ↑ dispersion under by “m” type and “c” type
 - No direct data
 - Proxy for “home” airport: ENDDOMO (carrier’s share of originating passengers at endpoint airports)

Adjusting for Cost Differences

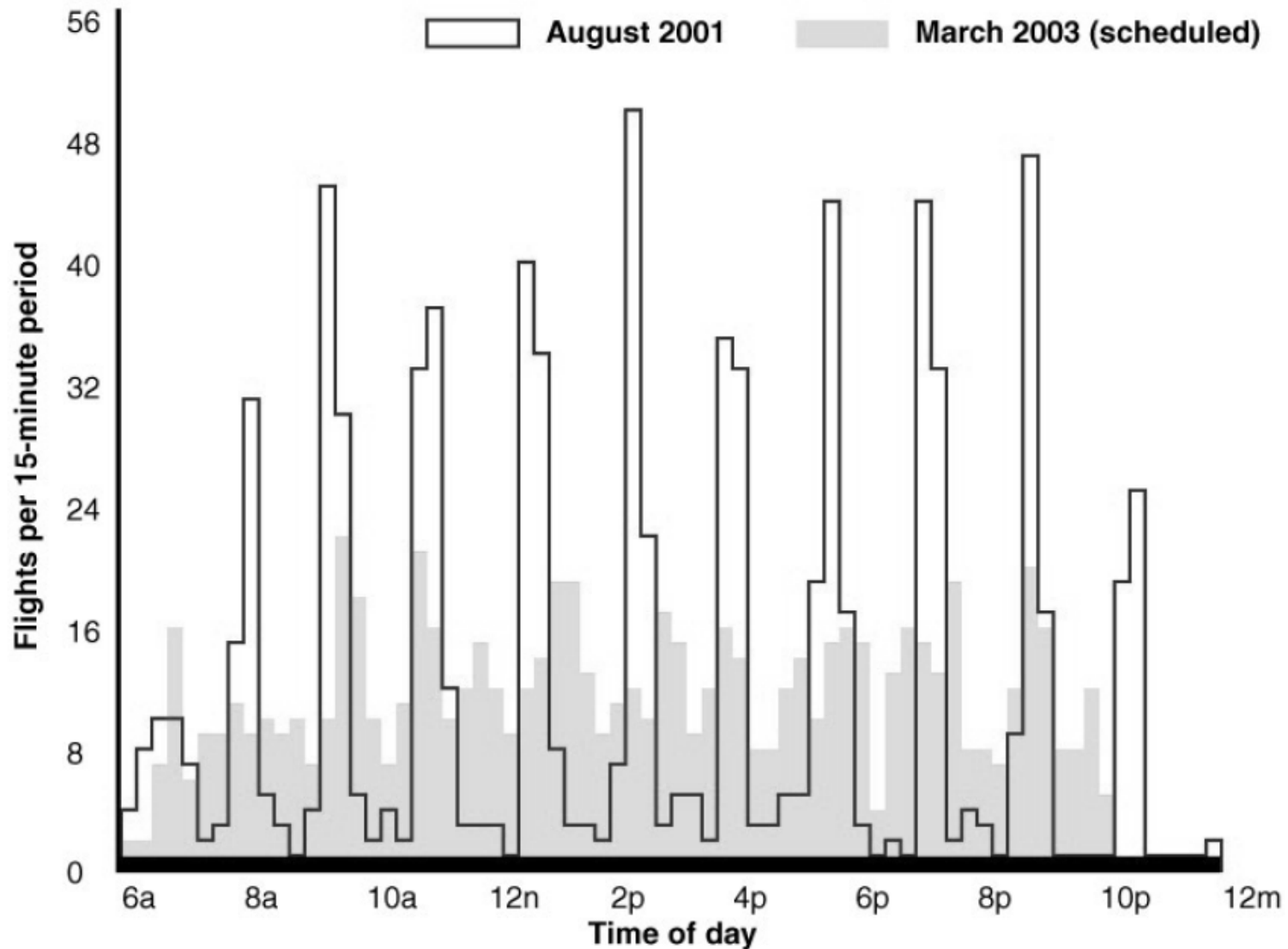
- Stigler (1987): firm price discriminates when ratio of prices is different from ratio of marginal (opportunity) costs
 - Cannot measure MC for airlines (think OR problem)
- Peak-load pricing (due to capacity constraints, prices need to adjust to divert passengers from peak to off-peak times)
 - “Systematic”: demand differences that are expected at the time flights are scheduled
 - “Stochastic”: uncertainty in demand resolved after schedule is determined

Measuring Systematic Peak-load Pricing

- Daily or weekly peaks in aircraft usage or in runway usage
 - BUT, they don't have data on this! (for a specific itinerary)
- More variation in a given carriers utilization (e.g. sometimes empty, sometimes full) likely to generate from cost-based price dispersion
- Shadow costs are likely to be highly nonlinear functions of fleet (and airport) utilization
- Proxy:
 - SDCAPFLT: For each carrier's "flight" (time of day, day of week), find avg fleet utilization rate, cube it, and find stdev across all flights
 - SDCAPAPT: For 22 congested airports, stdev of cubed airport utilization rate for each carrier's "flights". For other airports, 0.

“Banking” at Hub Airports

American at DFW



Source: Borenstein & Rose (2007)

Measuring Stochastic Peak-load Pricing

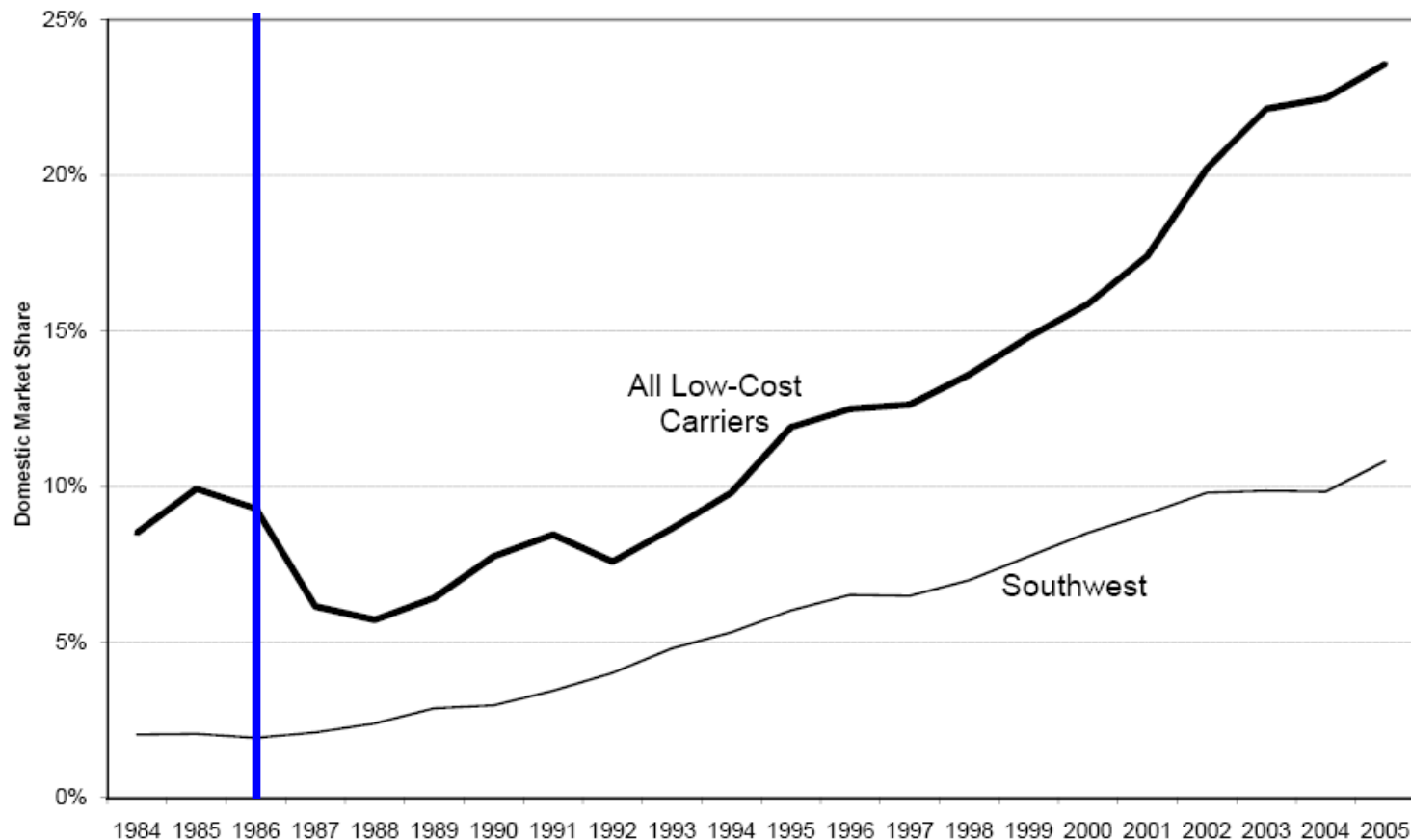
- Can't do anything without flight-level data on prices and utilization ("load factor")
- Again, see Puller, Sengupta, Wiggins (2012)
 - We test if metrics of scarcity drive intra-route price dispersion

Data

- 1986 Q2 DOT's Origin & Destination Survey
 - b/c combines entire quarter, include a measure of variation in lowest fare across 3 months INQGINI
 - Does temporal dispersion drive the dispersion?
 - Coach class direct or nonstop service
 - 1201 largest airport pairs where >80% traveled without change of plane
 - 11 major carriers: American, Continental, Delta, Eastern, Northwest, Piedmont, Republic, TWA, United, Western, USAir
 - Note: No low-cost carriers (keep in mind when we get to Gerardi & Shapiro)
- ➔ 1021 carrier-route observations on 521 routes

Historical Perspective – Size of Legacy vs. Low-Cost Carriers

Figure 7: Domestic Market Share of Southwest and All Low-Cost Carriers, 1984-2005



Source: Borenstein & Rose (2008)

Basic Empirical Specification

$$\begin{aligned}\ln \text{GINI}_{ijk} = & \beta_0 + \beta_1 \ln \text{FLTHERF}_{ij} \\ & + \beta_2 \ln \text{FLTSHARE}_{ijk} + \beta_3 \ln \text{FLTTOT}_{ij} \\ & + \beta_4 \ln \text{ENDDOMO}_{ijk} + \beta_5 \ln \text{TOURIST}_{ijk} \\ & + \beta_6 \ln \text{SDCAPFLT}_{ijk} + \beta_7 \ln \text{SDCAPAPT}_{ijk} \\ & + \beta_8 \text{DUMAPT}_{ij} + \beta_9 \ln \text{INQGINI}_{ijk} \\ & + \beta_{10} \text{DUMINQ}_{ijk} + \alpha_k + \gamma_{ij} + \eta_{ijk},\end{aligned}$$

k=carrier, i=airport 1, j=airport 2

Possible endogeneity: FLTTOT, FLTSHARE, FLTHERF

Use IV

Potential Endogeneity of FLTTOT, FLTSHARE, FLTHERF

- Density (FLTTOT) – dispersion may attract entry
 - IV = DISTANCE_{ij}
 - G&S will say this IV is problematic
 - Also IVs = means of population of airports i and j (using arithmetic and geometric means)
 - PAX1975 – total passengers on route 1975
- FLTSHARE – pricing may affect share
 - IV = (geometric) mean of share of enplanements at i and j
- FLTHERF (HHI based on FLTSHARE)

- IV = XFLTHERF: An instrument for FLTHERF assuming that the concentration of the flights on a route that are not performed by the observed airline is exogenous with respect to the price of the observed carrier, for example, that TWA's price on the Kennedy Airport–Los Angeles route does not affect the division of non-TWA flights between American and United. XFLTHERF is the square of a fitted value for FLTSHARE (from its first-stage regression) plus the “rescaled” sum of the squares of all other carriers' shares:

$$\text{XFLTHERF} = \widehat{\text{FLTSHARE}}^2 + \frac{\text{FLTHERF} - \text{FLTSHARE}^2}{(1 - \widehat{\text{FLTSHARE}})^2} \cdot (1 - \widehat{\text{FLTSHARE}})^2.$$

VARIABLE	PRICE DISCRIMINATION			PEAK-LOAD PRICING: Systematic and Stochastic (4)
	Monopoly Markets: Monopoly- Type (1)	Nonmonopoly Markets Monopoly- Type (2)	Competitive- Type (3)	
Concentration (FLTHERF)	+	+	-	?
Market density (FLTTOT)	+	-	-	?
Market share (FLTSHARE)	+	+	-	?
Endpoint dominance (ENDDOMO)	+	+	+	0
Population homogeneity (TOURIST)	-	-	-	?
Variation in shadow flight ca- pacity cost (SDCAPFLT)	0	0	0	+
Variation in shadow airport capacity cost (SDCAPAPT)	0	0	0	+
Uncongested airport (DUMAPT)	0	0	0	-

Monop $\equiv$ 1 carrier > 90% flights
 Duop $\equiv$ 2 carriers cumulatives > 90%
 Competitive $\equiv$ others

TABLE 3
PRICE DISPERSION REGRESSION RESULTS
(Dependent Variable: GINI; $N = 1,020$)

VARIABLE	MODEL 1: PARSIMONIOUS		MODEL 2: MARKET STRUCTURE	
	Log-Log (1)	Linear (2)	Log-Log (3)	Linear (4)
CONSTANT	-1.742 (.039)	.197 (.005)		
MONOPOLY			-2.169 (.411)	.154 (.032)
DUOPOLY			-2.033 (.215)	.174 (.035)
LARGE-DUOP			-.117 (.055)	-.022 (.010)
SMALL-DUOP			-.067 (.061)	-.017 (.009)
COMPETITIVE			-1.807 (.259)	.172 (.024)
FLTHERF	-.323 (.072)	-.092 (.026)		
FLTHERF-COMP			-.250 (.126)	-.141 (.064)
FLTTOT	-.169 (.052)	-.036 (.035)		
FLTTOT-MONOP			-.178 (.136)	-.268 (.289)
FLTTOT-DUOP			-.241 (.079)	-.067 (.102)
FLTTOT-COMP			-.041 (.063)	.009 (.038)
FLTSHARE	.010 (.057)	.033 (.026)		
FLTSHARE-COMP			.049 (.052)	.104 (.029)
ENDDOMO	.087 (.028)	.063 (.023)	.075 (.026)	.035 (.021)
TOURIST	-.066 (.023)	-.549 (.173)	-.055 (.023)	-.483 (.176)
SDCAPFLT	.061 (.027)	-.012 (.033)	.053 (.023)	-.031 (.030)
SDCAPAPT	.032 (.021)	.070 (.039)	.027 (.018)	.045 (.038)
DUMAPT	-.345 (.078)	-.023 (.009)	-.316 (.071)	-.024 (.009)
INQGINI	.047 (.014)	.091 (.030)	.043 (.014)	.094 (.030)
INQDUM	-.154 (.050)		-.144 (.050)	

NOTE.—All regressions include carrier-specific effects. Asymptotic standard errors are in parentheses.

Is There Other Stuff Going On?

- Regressions explain only 20% of dispersion
- Airlines may price differently, say, if they own a “yield management system”

ESTIMATED CARRIER-SPECIFIC EFFECTS FROM TABLE 3 REGRESSIONS
(Dependent Variable: GINI; $N = 1,020$)

	MODEL 1: PARSIMONIOUS		MODEL 2: MARKET STRUCTURE	
	Log-Log	Linear	Log-Log	Linear
Owners of computer reservation systems:				
Delta	-.016 (.037)	-.011 (.006)	-.020 (.037)	-.011 (.006)
Eastern	-.075 (.039)	-.016 (.006)	-.088 (.039)	-.019 (.006)
TWA	.244 (.049)	.043 (.008)	.262 (.049)	.046 (.008)
United	-.135 (.037)	-.024 (.006)	-.138 (.037)	-.024 (.006)
Nonowners of computer reservation systems:				
Continental	-.166 (.045)	-.033 (.007)	-.163 (.046)	-.032 (.007)
Northwest	-.148 (.047)	-.033 (.007)	-.144 (.047)	-.032 (.007)
Piedmont	.023 (.049)	-.008 (.008)	.014 (.046)	-.011 (.007)
Republic	-.116 (.047)	-.026 (.007)	-.110 (.047)	-.026 (.008)
USAir	-.165 (.045)	-.034 (.007)	-.163 (.045)	-.034 (.007)
Western	-.141 (.066)	-.029 (.010)	-.163 (.066)	-.034 (.010)

→ Owners of CRSs have
More dispersion on avg.

NOTE.—All effects are estimated relative to American Airlines. Asymptotic standard errors are in parentheses.

B&R's Conclusions

- “The findings indicate that traditional monopoly theories of price discrimination may give neither complete nor accurate predictions of pricing patterns in monopolistically competitive markets.”

Gerardi & Shapiro Follow-Up

Use panel of routes. Exploit time series variation in concentration (e.g. exit, entry)

TABLE 2
PANEL ESTIMATES
Dependent Variable: G_{ijt}^{odd}

	A. ALL ROUTES			B. BIG-CITY ROUTES			C. LEISURE ROUTES		
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
$-\ln \widehat{\text{HERF}}$	-.121*** (.022)			-.229*** (.038)			-.076* (.044)		
$\ln \hat{N}$		-.177*** (.029)			-.357*** (.060)			-.132** (.058)	
N^{LEG}			-.005 (.004)			-.013** (.006)			-.006 (.009)
N^{LCC}			-.056*** (.007)			-.089*** (.010)			-.034*** (.012)
Observations	112,499	112,499	112,499	32,188	32,188	32,188	24,555	24,555	24,555

NOTE.—All regressions include carrier-route-specific dummies, time dummies, quarter dummies, and a dummy variable indicating whether the carrier is in bankruptcy. Standard errors are in parentheses and are clustered by route to account for both autocorrelation and correlation between carriers on the same route.

Note: they switch signs on HERF ($\uparrow -\ln \text{HERF} \rightarrow$ more competition)

G&S's Characterization of Theory

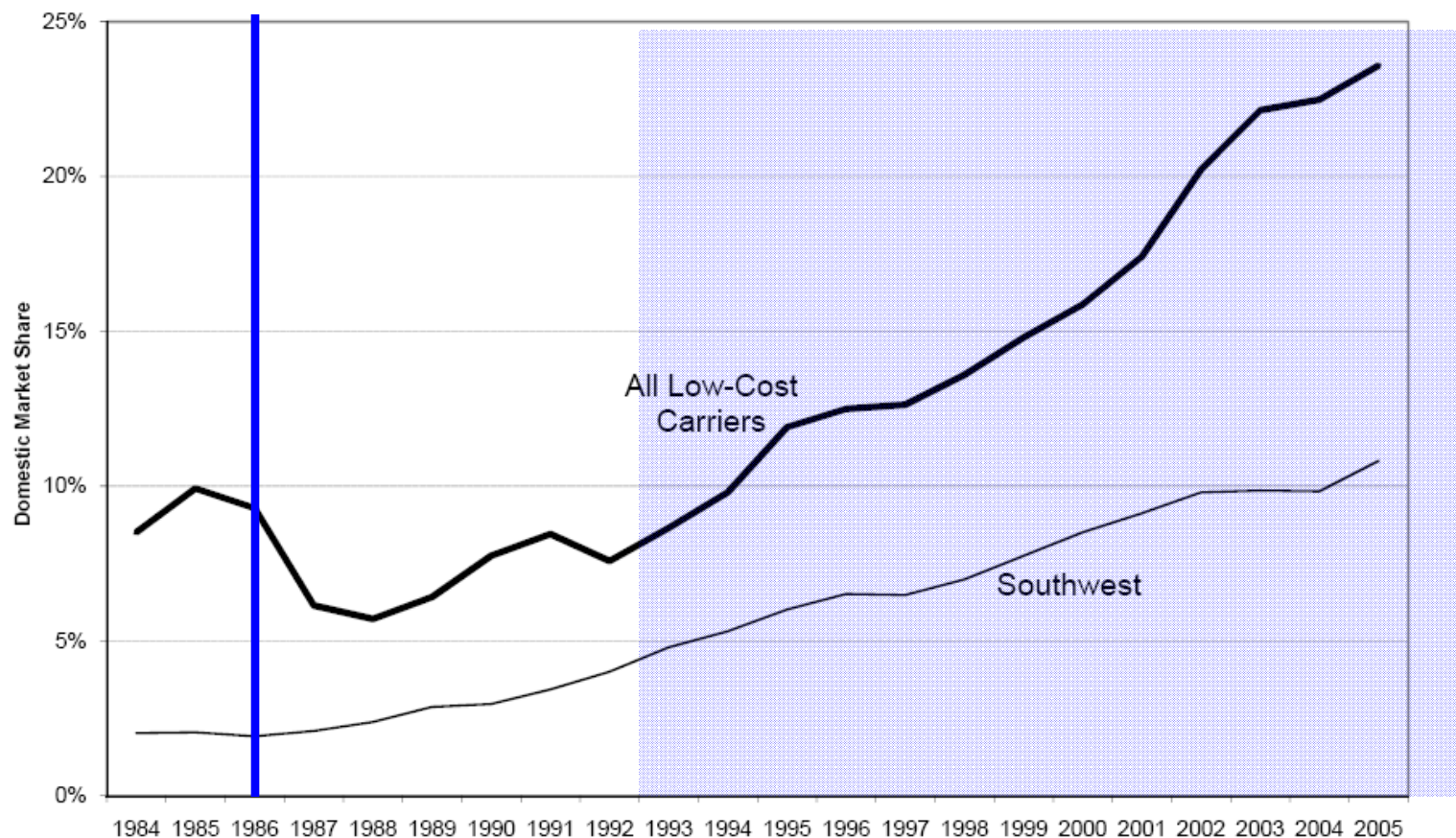
- “Textbook” price discrimination
 - As competition increases, it's more difficult to $P > MC$, so competition makes it more difficult to price discriminate
- B&R's “brand loyalty”
 - Competition will have less effect on prices charged to high-fare, business travelers (due to Frequent Flyer Programs) than on prices charged to non-brand loyal leisure travelers
 - I.e. reduces prices to both, but reduces low prices more. Hence, competition increases dispersion.
 - Reinterpretation of “monopoly-type” and “competitive-type” price discrimination

G&S Sample

- 1993:Q1 to 2006:Q3
 - 9 major carriers plus LCCs and regionals
- ➔ the industry G&S study is very different from the one that B&R study!

Historical Perspective – Size of Legacy vs. Low-Cost Carriers

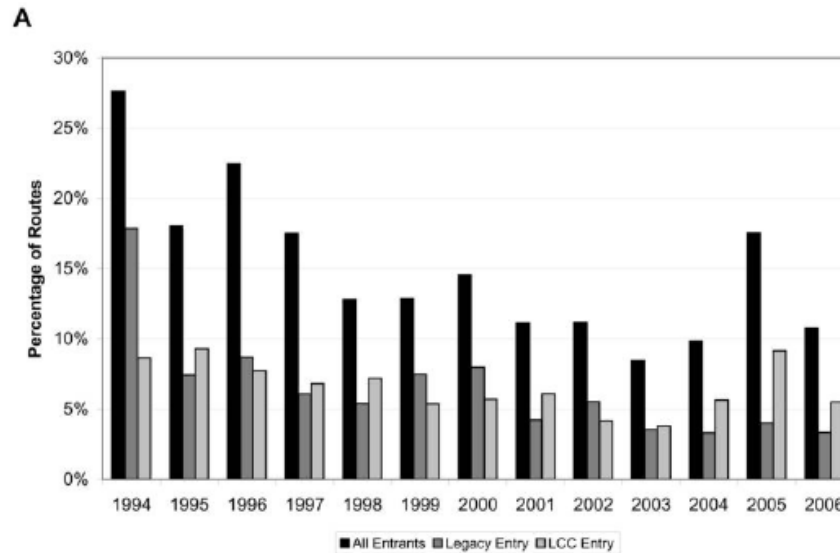
Figure 7: Domestic Market Share of Southwest and All Low-Cost Carriers, 1984-2005



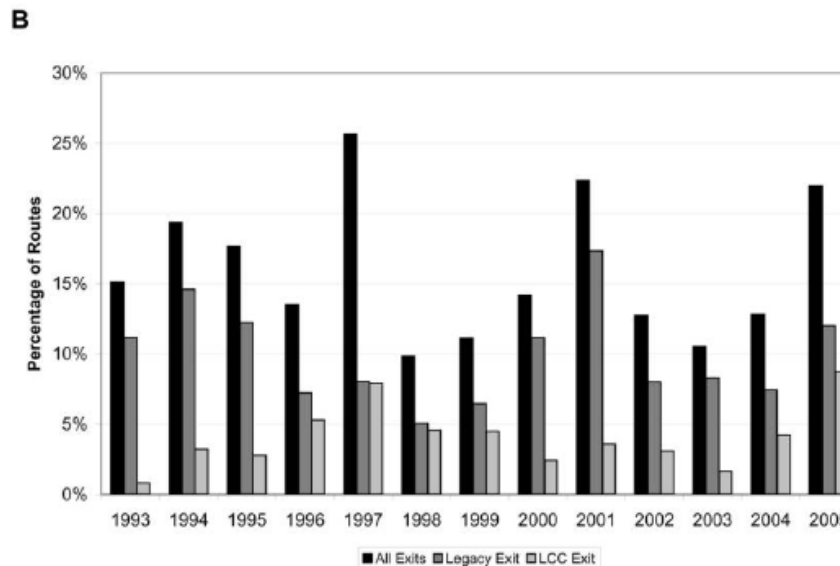
Source: Borenstein & Rose (2008)

Variation in Competition

Entry:



Exit:



Isolating Price Discrimination

- Causes of dispersion
 - “textbook” price discrimination
 - Systematic and stochastic peak-load pricing
- Assert that routes with more heterogeneity in customer type will have differing dispersion due to P.D. but not due to peak-load pricing
 - Assume demand uncertainty does not change with heterogeneity (Dana 1999 model)
- So “leisure” routes are mostly elastic customers; “big city” routes are mix of elastic and inelastic
 - “leisure” $\equiv$ top 15% of cities measured by accommodation/nonfarm earnings
 - “big city” $\equiv$ both origin & dest in one of 30 largest MSAs

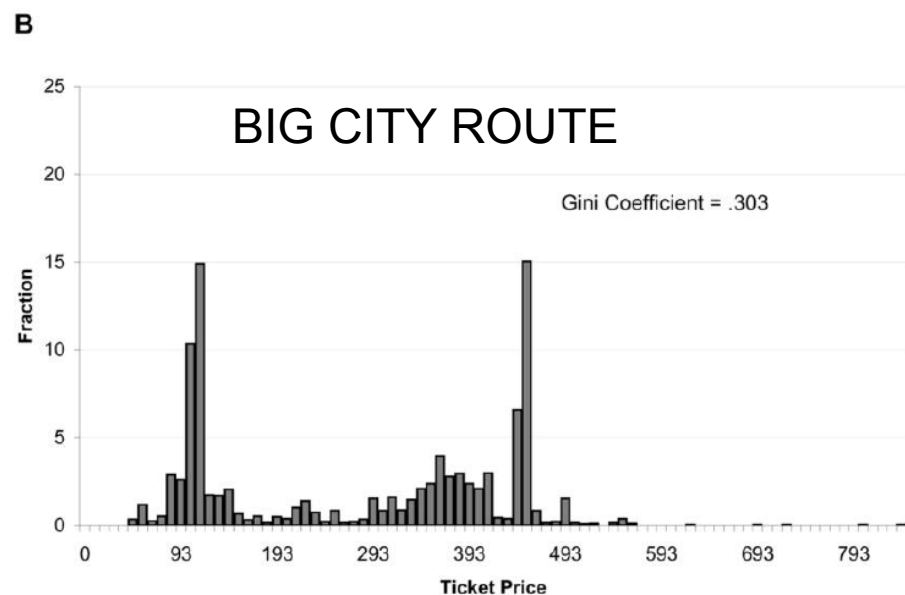
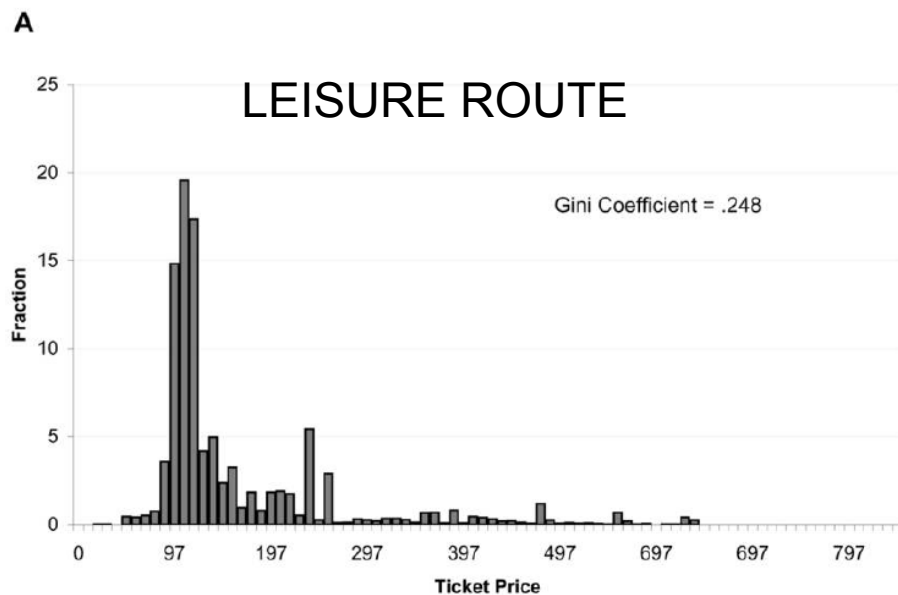


TABLE 1
MEDIAN PRICE DISPERSION BY SUBSAMPLE

	Big-City Routes (1)	Leisure Routes (2)	All Routes (3)
Monopoly routes	.29	.19	.23
Competitive routes	.25	.18	.21
All routes	.26	.18	.22

NOTE.—This table reports the median Gini coefficient of domestic coach-class fares for different samples of the data. Monopoly routes are those in which the average market share over the sample period is greater than 0.95 for a single carrier.

Suggestive that competition has bigger “effect” on dispersion for Big-City routes

FIG. 2.—Example price distributions: histograms of coach-class fares during the first quarter of 1999. *A*, Leisure route, Philadelphia (PHL) to Orlando (MCO) on US Airways. *B*, Big-city route, Philadelphia (PHL) to Chicago (ORD) on United Airlines. Prices are in nominal U.S. dollars and are computed as directional fares (round-trip fares are divided by 2).

Empirical Specification

$$G_{ijt}^{\text{lodd}} = \alpha + \beta \times \text{Competition}_{jt} + \theta \times X_{it} + \gamma_t + v_{ij} + \nu_{ijt},$$

- i =carrier, j =route
- U_{ij} = carrier-route fixed effects (which may capture confounders in B&R)
- “cluster” std error on route
- How do we interpret?

TABLE 2
PANEL ESTIMATES
Dependent Variable: G_{ijt}^{lodd}

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* Significant at the 10 percent significance level.

** Significant at the 5 percent significance level.

*** Significant at the 1 percent significance level.

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$$G_{ijt}^{\text{lodd}} = \alpha + \beta \times \text{Competition}_{jt} + \theta \times X_{it} + \gamma_t + v_{ij} + \nu_{ijt},$$

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Which Types of Fares Fall More?

TABLE 3
PANEL ESTIMATES
Dependent Variable: Log of 90th and 10th Percentiles

	BIG-CITY ROUTES						LEISURE ROUTES					
	ln P90			ln P10			ln P90			ln P10		
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
$-\ln \widehat{\text{HERF}}$	-.406*** (.041)			-.166*** (.027)			-.168*** (.028)			-.119*** (.024)		
$\ln \hat{N}$		-.652*** (.067)			-.273*** (.039)			-.259*** (.041)			-.176*** (.036)	
N^{LEG}			-.041*** (.005)			-.025*** (.004)			-.011 (.008)			-.008 (.007)
N^{LCC}			-.156*** (.014)			-.054*** (.007)			-.090*** (.011)			-.065*** (.009)
Observations	32,188	32,188	32,188	32,188	32,188	32,188	24,555	24,555	24,555	24,555	24,555	24,555

NOTE.—All regressions include carrier-route-specific dummies, time dummies, and a dummy variable indicating whether the carrier is in bankruptcy. Hats indicate that instrumental variables were used. Standard errors are in parentheses and are clustered by route to account for both autocorrelation and correlation between carriers on the same route.

* Significant at the 10 percent significance level.

** Significant at the 5 percent significance level.

*** Significant at the 1 percent significance level.

Reconciling B&R with G&S

- Possible explanations
 - Industry different in 1986 (B&R) vs. '93-'06 (G&S)
 - Do G&S results hold for routes “similar” to industry in 1986 (i.e. major-dominated routes)? They don't say
 - B&R suffer from “omitted variable bias”
 - In their cross-section, is there a major determinant of price discrimination that correl w/ concentration
 - G&S control for unobserved effects for a carrier-route that don't vary over time

Can They Replicate B&R Using the Cross-Section?

$$G_{ij}^{\text{odd}} = \beta_0 + \beta_1 \ln \widehat{\text{MKTSHARE}}_{ij} - \beta_2 \ln \widehat{\text{HERF}}_{ij} + \beta_3 \ln \widehat{\text{FLTOT}}_j + \beta_4 \ln \text{TOURIST}_j + \beta_5 \text{HUB}_{ij} + \beta_6 \text{SMALL}_j + \alpha_i + \gamma_j + \eta_{ij}. \quad (3)$$

TABLE 4
CROSS-SECTIONAL ESTIMATES
Dependent Variable: G_{ij}^{odd}

	1993:Q1		1999:Q1		2005:Q1	
	(1)	(2)	(1)	(2)	(1)	(2)
$-\ln \widehat{\text{HERF}}$	.069 (.055)		.337*** (.058)		.179*** (.042)	
$\ln \widehat{\text{MKTSHARE}}$	.047*** (.015)		.119*** (.017)		.101*** (.015)	
$\ln \hat{N}$		.164*** (.038)		.244*** (.039)		.035 (.029)
$\ln \widehat{\text{FLTOT}}$	-.036*** (.013)	-.074*** (.013)	-.041*** (.013)	-.084*** (.014)	-.010 (.010)	-.021** (.010)
$\ln \text{TOURIST}$	-.020 (.014)	-.038*** (.014)	-.043*** (.013)	-.059*** (.013)	-.030*** (.009)	-.029*** (.009)
HUB	.054* (.028)	.140*** (.026)	.133*** (.034)	.320*** (.029)	.108*** (.027)	.208*** (.024)
SMALL	-.129*** (.034)	-.067* (.037)	-.191*** (.034)	-.149*** (.036)	-.058** (.025)	-.061** (.025)
Observations	1,993	1,993	2,085	2,085	2,053	2,053

NOTE.—All regressions include carrier-specific dummies. Route-specific effects are considered random in these regressions. Hats indicate that instrumental variables were used. Robust standard errors are in parentheses.

* Significant at the 10 percent significance level.

** Significant at the 5 percent significance level.

*** Significant at the 1 percent significance level.

Yes, coefficients of $-\ln \text{HERF} > 0$ in 53 of 55 quarters

Potential Confounder

- $\text{Corr}(\text{IV}, u_j)=0?$ $\text{Corr}(\text{DIST}, u_j)=0?$
- Claim: plane size is correl w/ distance
 - Small planes not fuel efficient for long routes
- Claim: plane size is correl w/ u_j – ability to price discriminate
 - “Price dispersion is also likely related to the size of the plane, since larger planes contain more seats and thus provide an airline with a greater opportunity to implement strategies that create more price dispersion.”
 - Issues:
 - How much variation in plane size was there in 1986 for majors??
 - Is this driven by regional airlines in G&S sample period?

Effect of Adding Plane Size Variable

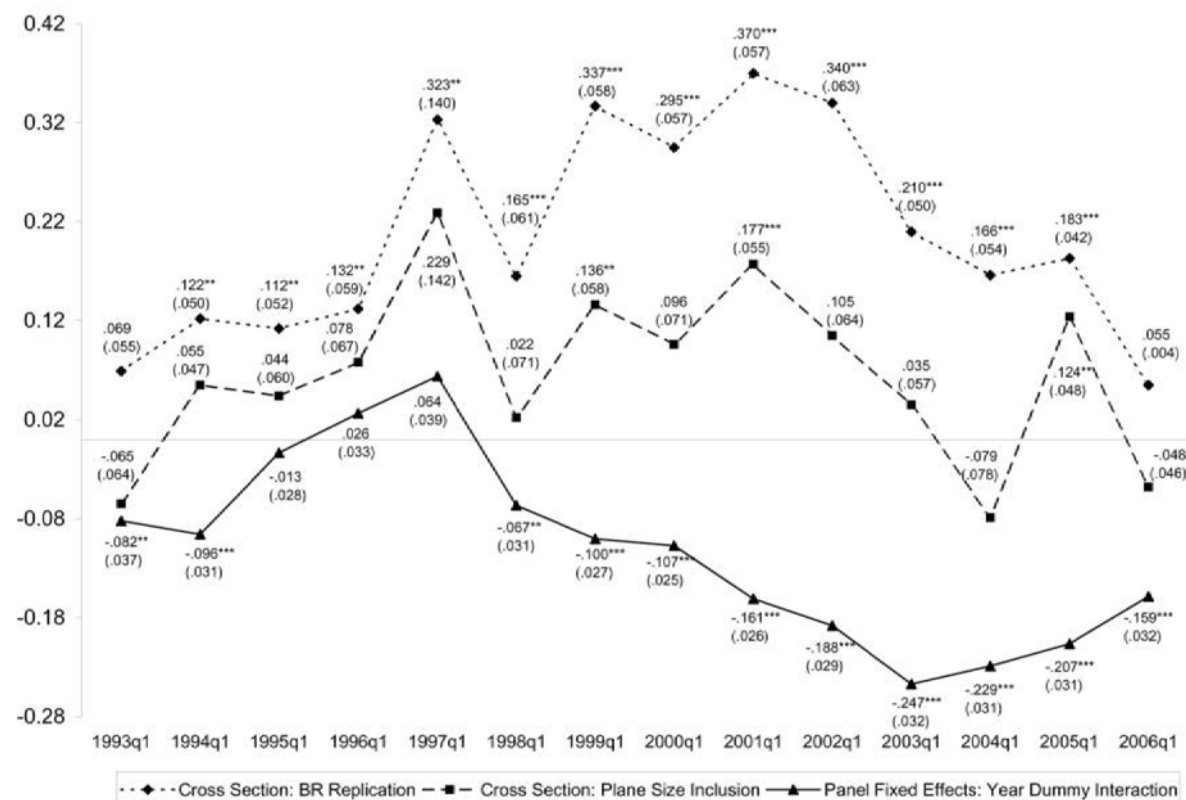
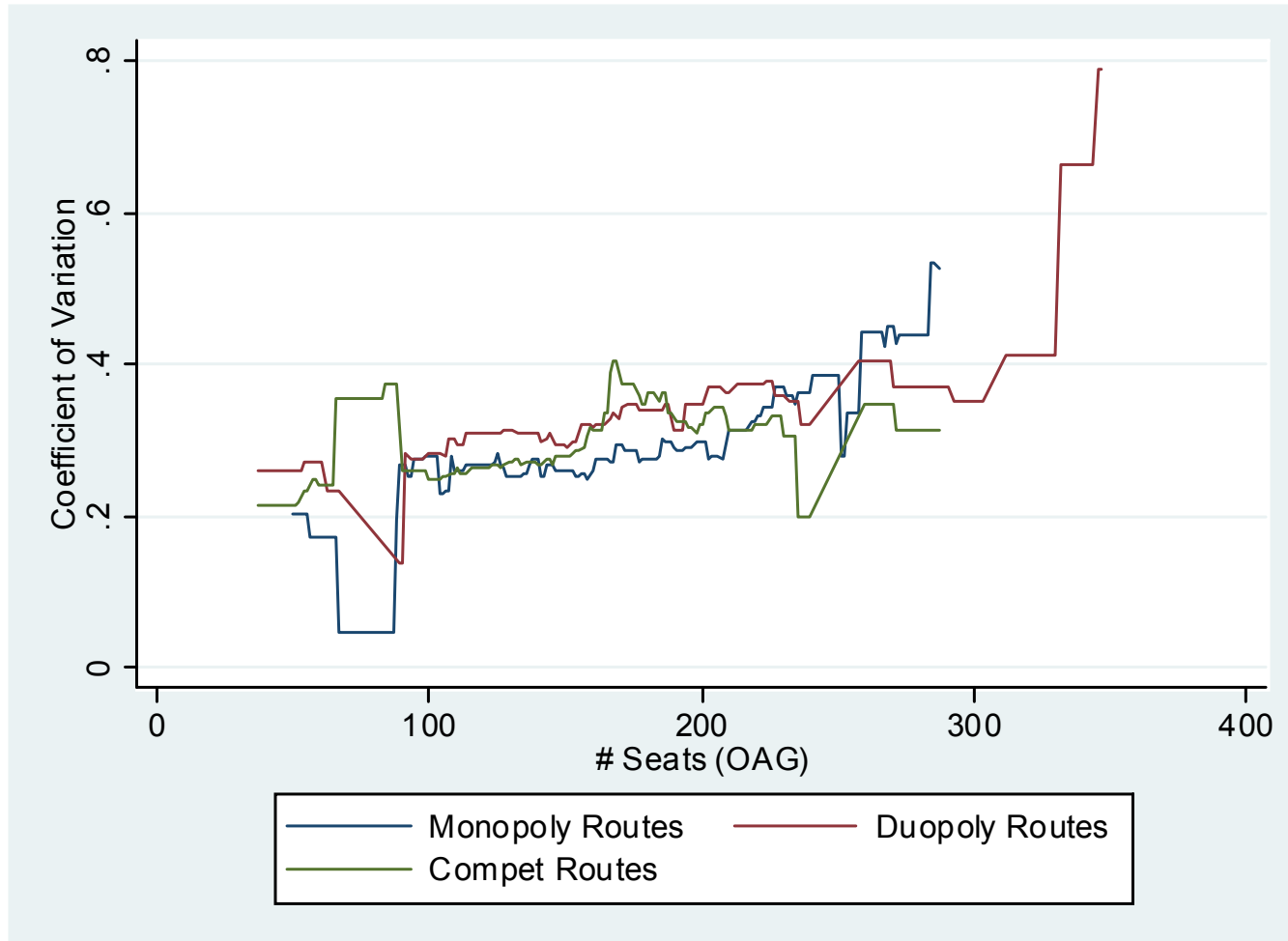


FIG. 4.—The effects of competition on price dispersion. The figure shows estimates (for three different regressions) of the coefficient on competition as measured by $-\ln \widehat{HERF}$. Diamonds indicate estimates of $-\ln \widehat{HERF}$ under equation (4) for quarter 1 of each year in our sample, squares indicate estimates of the same equation including average seat capacity as an endogenous explanatory variable, and triangles indicate estimates on the year-dummy interactions of the fixed-effects panel analysis. Robust standard errors are in parentheses. One, two, and three asterisks indicate significance at the 10 percent, 5 percent, and 1 percent significance level, respectively.

Do Bigger Planes Have More Price Dispersion (After Controls)?



Source: data used in Puller, Sengupta, Wiggins

Do Potential Fencing Devices Drive Prices More in Big City Routes?

- Consider model with “textbook” P.D.
 - Need fencing device to induce customers to self-select to “leisure” or “business” (or somewhere in between)
 - Possibilities:
 - Refundability
 - Restrictions on days of week that travel can occur
 - Minimum or maximum stay restrictions (e.g. must stay 1 night)
 - Stay over a Saturday night
 - Are these **associated** with different fares?
 - Need ticket restriction data. DB1B does not have. Puller, Sengupta, Wiggins dataset does.

Baseline: Across All Routes

Dependent Variable: Log(Fare)		
	(1)	(2)
	Characteristics Only	Actual LF
Advance_0_3	0.292 (0.011)**	0.295 (0.011)**
Advance_4_6	0.262 (0.011)**	0.265 (0.011)**
Advance_7_13	0.180 (0.008)**	0.182 (0.008)**
Advance_14_21	0.056 (0.008)**	0.058 (0.008)**
Refundable	0.497 (0.009)**	0.497 (0.009)**
Roundtrip Itinerary	-0.116 (0.004)**	-0.117 (0.004)**
Travel Restriction	-0.304 (0.004)**	-0.302 (0.004)**
Stay Restriction	-0.080 (0.005)**	-0.081 (0.005)**
Stayed Over Saturday Night	-0.131 (0.006)**	-0.126 (0.006)**
LF_Actual - Averaged across flight segments		0.045 (0.005)**
LF_Expected - Averaged across flight segments		
LF_Actual - Maximum across flight segments		
LF_Expected - Maximum across flight segments		
Observations	224,108	224,108
R-squared	0.695	0.696

Split By Big City vs. Leisure

Dependent Variable: Log(Fare)

	<u>BIG CITY - MONOPOLY</u>	<u>BIG CITY - COMPETITIVE</u>	<u>LEISURE - MONOPOLY</u>	<u>LEISURE - COMPETITIVE</u>
	Coef.	Coef.	Coef.	Coef.
Advance_0_3	0.35 *	0.19 *	0.45 *	0.11 *
Advance_4_6	0.31 *	0.15 *	0.48 *	0.17 *
Advance_7_13	0.17 *	0.12 *	0.38 *	0.18 *
Advance_14_21	0.04 *	0.01	0.13 *	0.12 *
Refundable	0.90 *	0.46 *	0.05	1.06 *
Roundtrip Itinerary	-0.04 *	-0.12 *	-0.16 *	-0.32 *
Travel Restriction	-0.20 *	-0.33 *	-0.13 *	-0.17 *
Stay Restriction	0.01	-0.05 *	-0.27 *	0.01
Stayed Over Saturday Night	-0.23 *	-0.11 *	0.01	0.03
Load Factor Actual	0.00	-0.01	0.02	0.05 *
Load Factor Expected	0.15 *	0.12 *	0.05	0.02
N	25610	75558	5714	12333

Reconciling?

- If you can reconcile convincingly, that would be a nice paper